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Abstract. Despite well-documented vulnerabilities, passwords remain the dominant method of user authentication. Passkeys offer a cryptographically robust alternative, backed by major platform vendors. Yet adoption remains modest, suggesting that barriers to uptake are shaped not only by technical considerations but by subjective user perceptions that conventional adoption models may not fully capture. This study employs Q-methodology to systematically elicit and characterise the viewpoints underpinning passkey non-adoption among daily technology users. We recruited 40 participants to complete an online Q-sort and share their perceptions of passkeys. Four distinct viewpoints emerged: satisfaction with existing password-based practices augmented by password managers and MFA; cautious optimism toward passkeys tempered by concerns about single points of failure and loss of control; frustration with fragmented user experiences; and uncertainty rooted in limited technical understanding of passkey mechanisms. These findings suggest that passkey non-adoption is not due to a single barrier but reflects a heterogeneous landscape of viewpoints, each requiring distinct intervention strategies.

1 Introduction

Notwithstanding regular criticism and frequent practical exposure, passwords continue to be the predominant method of user identification [12]. Users frequently choose weak or predictable passwords, or reuse them across multiple systems, making them highly susceptible to brute-force attacks, credential stuffing, and phishing [11]. Even when strong passwords are used, the cognitive burden of remembering them often leads to insecure practices, such as writing them down or relying on simplistic patterns.

Bonneau et al. [3] uses three categories to compare web authentication mechanisms: usability, deployability, and security. Passwords score high in the deployability category, but less so in the usability and security categories. While it is easy to learn how to use passwords, it does take significant effort to remember them. Given the frequency of password compromises, it is advisable to have one

password for each system/service, a task which is not scalable or within the cognitive abilities of users using tens, or even hundreds of systems.

In response, the security community increasingly advocates for mechanisms such as multi-factor authentication (MFA), password managers, and emerging passwordless technologies, such as passkeys and biometrics. Among these alternatives, passkeys — a passwordless authentication mechanism built on the FIDO2 and WebAuthn standards [9] — have emerged as a compelling proposition. Passkeys offer a unique combination of security properties. In addition, major platform vendors — for example, Apple, Google, and Microsoft — have integrated passkey support into their operating systems and browsers, signalling strong industry commitment to the technology [14].

However, technical merit and industry endorsement do not, in themselves, guarantee user adoption. Despite the availability of passkey support across major platforms since 2022, evidence suggests that uptake remains modest [14,8,10]. This disparity between availability and adoption suggests that the barriers to passkey uptake are not purely technical but are shaped by subjective user perceptions, attitudes, and concerns that existing quantitative adoption models may not fully capture. Thus the primary research question driving this study: *"Why are users not adopting passkeys?"*

To explore this question, we employ Q-methodology, an approach designed to systematically elicit and categorise the subjective viewpoints that underpin non-adoption.

2 Background

2.1 Passkeys

Passkeys employ asymmetric key pairs, with a private key securely stored on the user's device and a corresponding public key held by the relying party i.e., the entity that needs to verify the user's identity. Authentication occurs through a cryptographic challenge-response protocol, typically gated by a local biometric or device PIN, thereby eliminating the transmission of shared secrets.

In terms of the Bonneau et al. framework [3], passkeys represent a marked improvement over passwords in both the usability and security categories, while their growing platform integration increasingly addresses earlier deployability concerns.

Regarding security, the use of asymmetric cryptography eliminates the transmission and storage of shared secrets, rendering credential phishing, stuffing, and server-side breaches ineffective as attack vectors. The cryptographic binding of each assertion to the relying party's origin domain provides structural phishing resistance that does not depend on user vigilance.

In terms of usability, passkeys replace the cognitive burden of password creation, memorisation, and management with a single biometric gesture or device unlock — an interaction pattern most users already perform dozens of times daily. However, passkeys do not yet achieve a clean sweep across all of Bonneau

et al.'s usability properties: questions around account recovery, the mental model required to understand what a passkey is, and cross-platform operability remain open.

On deployability, passkeys benefit from the backing of major platform vendors and the absence of any requirement for specialised hardware beyond a modern smartphone, though relying parties must still implement WebAuthn support and navigate the complexities of a fragmented ecosystem of credential synchronisation providers. The deployability challenges are particularly acute in enterprise settings, where Kepkowski et al. [13] found that enterprise FIDO2 deployment is impeded not by fundamental technical shortcomings but by practical barriers including insufficient implementation know-how, inadequate account recovery tooling, high costs, limited native integration with enterprise infrastructure, and inconsistent user experience across platforms.

2.2 Technology Adoption and Resistance

Research into how individuals engage with new technologies has evolved along three broadly complementary trajectories. The first, rooted in Davis's Technology Acceptance Model (TAM) [7] and its successor Unified Theory of Acceptance and Use of Technology (UTAUT) [24], foregrounds individual perceptions — principally perceived usefulness, ease of use, and social influence — as predictors of adoption intention. These variance-based models have dominated the information systems literature for over three decades and remain widely applied, though they have been critiqued for their confirmatory orientation and reliance on predefined constructs.

The second trajectory draws on Rogers's Diffusion of Innovations (DOI) theory [21], which situates adoption within a broader social system and emphasises how attributes of the innovation itself — relative advantage, compatibility, complexity, trialability, and observability — shape the rate and pattern of diffusion across populations.

The third trajectory shifts attention from adoption to resistance. Ram and Sheth's Innovation Resistance Theory (IRT) [18] argues that non-adoption is not merely the absence of positive drivers but an active response shaped by functional barriers — relating to usage, value, and risk — and psychological barriers arising from tradition and image conflicts.

All three perspectives have been applied to technology adoption broadly, yet none has been systematically applied to the specific question of passkey non-adoption. Collectively, these three perspectives suggest that understanding technology uptake requires attention to both what facilitates adoption and what actively impedes it. These frameworks also share a limitation — they treat the object of adoption as a technology artefact rather than as a service.

2.3 Authentication as a Service: A Value Perspective

The preceding discussion examined passkey non-adoption through the lens of technology acceptance and innovation resistance frameworks. These perspectives

offer well-established constructs for understanding individual adoption decisions, yet they share a common assumption: that authentication is a technology artefact to be accepted or rejected. We propose a complementary framing that treats authentication not merely as a technology, but as a service consumed by users — and one whose adoption is therefore shaped by the value users perceive it to deliver.

This perspective draws on the IT Service Management (ITSM) literature, and specifically on the Information Technology Infrastructure Library (ITIL) framework's articulation of service value. According to ITIL [6], the value of a service is derived from a combination of utility and warranty. Utility refers to fitness for purpose — the functionality a service offers to meet a particular need, either by supporting the consumer's performance or by removing constraints [6]. Warranty refers to fitness for use — the assurance that a service will perform adequately, typically encompassing availability, capacity, continuity, and security [6]. A service delivers value only when both utility and warranty are present in sufficient measure; a service that is fit for purpose but not fit for use fails to realise its intended value for the consumer.

Although the ITIL framework was developed primarily for organisational IT service delivery [15], its value model offers a useful analytical lens for understanding end-user perceptions of authentication mechanisms. Authentication, viewed as a service, exists to enable users to securely access systems and resources. Like any service, it is consumed repeatedly, its performance is experienced subjectively, and its value is assessed — consciously or otherwise — against the user's expectations of both what it does and how reliably it does it.

Applying this lens to passwords reveals an instructive asymmetry. From a user's perspective, passwords score reasonably well on utility: they are universally understood, function across virtually all platforms, afford users a sense of direct control, and permit flexible practices such as sharing credentials with trusted others. Their warranty, however, is poor — they are vulnerable to compromise, cognitively burdensome to manage at scale, and fragile in the face of the reuse and simplification strategies that users inevitably adopt [1,3,11]. Yet users have developed extensive coping mechanisms — password reuse hierarchies, personal algorithms, written records, and selective effort rationing [23] — that function, in service management terms, as informal workarounds that patch over warranty failures sufficiently for the service to remain, in the user's perception, adequate.

Passkeys, by contrast, offer a substantially improved warranty proposition: stronger cryptographic security, inherent phishing resistance, no shared secrets to be compromised, and no memorisation burden. However, their utility may be less clear to users who have not yet adopted them. Questions such as *"Does this work on all my devices?"*, *"What happens if I lose my phone?"*, *"Can I share access with a family member?"*, and *"Do I still have control over my credentials?"* represent utility concerns — doubts about whether the service does what the user needs it to do. Equally, users may harbour warranty concerns about passkeys themselves: *"Will this still work in five years?"*, *"What if the platform vendor changes its policy?"*, or *"Is my biometric data safe?"*. These concerns may coexist with, or

even outweigh, the warranty improvements that passkeys objectively offer over passwords.

The utility/warranty framing thus suggests that passkey non-adoption cannot be understood purely through constructs such as perceived usefulness and effort expectancy [24], nor solely through resistance categories such as usage barriers or risk barriers [18]. Users may be conducting an implicit value assessment that integrates both functional concerns (what the service does and whether it does it well enough) and comparative judgements against the incumbent service whose shortcomings they have already learned to manage. This is consistent with the broader ITSM insight that replacing an established service is not simply a matter of demonstrating technical superiority; the replacement must deliver recognisable value across both dimensions, and must do so in ways that are legible to the consumer [6].

Incorporating this service value perspective alongside the adoption and resistance frameworks discussed in Section 2.2 enriches the theoretical underpinnings of the study. Next we discuss the methodology.

3 Methodology

To examine user perceptions of passkeys, we employed Q-methodology (Q), a structured approach for systematically studying subjectivity by identifying and analysing shared viewpoints within a population [22]. Q has been adopted across multiple disciplines, including prior work in cybersecurity [19,17,2]. A key strength of this method is its ability to distinguish nuanced perspectives on a topic, offering richer insight than conventional Likert-scale surveys or the broad thematic patterns typically derived from interview-based research [16].

Q is a mixed-methods approach comprising a series of quantitative and qualitative stages. After establishing the research question, the next step is to construct the Q-sample — a collection of statements that reflect the range of discourse surrounding the topic. In Q the sample size is defined by the number of items in the Q-sample, making it essential to ensure that the statements adequately represent the breadth of relevant discourse [5]. For this study, we reviewed academic and grey literature and developed 24 statements that capture key issues related to passkey adoption. A subset of the discourse was selected to reduce the number of items presented for participant sorting while still capturing the breadth of discourse on the topic. The selection process followed Fisher's design of experiments, as recommended by Brown [4]. The statements addressed dimensions of utility and warranty, including functionality, availability, security, complexity, and compatibility, as defined by ITIL. The categorised statements are presented in Table 1. Participants sorted these statements into a researcher-provided fixed quasi-normal distribution, an activity referred to as a Q-sort, with item placement indicating the relative salience of each statement for the participant [16].

Table 1. Statements in the Q-sample.

Id	Statement	Category
1	I am happy with the status quo	Utility — Functionality
2	I think passwords are good enough	Utility — Functionality
3	I think password managers make password management less problematic	Utility — Functionality
4	I think Single Sign-on is good enough	Utility — Functionality
5	I must anyway use passwords on sites that support passkeys	Utility — Functionality
6	I am worried about the fallback mechanisms when passkeys don't work	Warranty — Availability
7	I am unsure how lost or stolen devices would be handled	Warranty — Availability
8	I don't know how new devices are handled	Warranty — Availability
9	I don't think my biometrics will stay private	Warranty — Security
10	I don't use biometrics	Warranty — Security
11	I don't trust providers	Warranty — Security
12	I prefer combining multiple methods (i.e. MFA)	Warranty — Security
13	I am unsure how multiple devices are handled	Warranty — Complexity
14	I find roaming authenticators hard to use	Warranty — Complexity
15	I find the setup of passkeys difficult	Warranty — Complexity
16	I think syncing across different platforms is hard	Warranty — Complexity
17	I find the terminology hard to understand	Warranty — Complexity
18	I find the user interface difficult to use	Warranty — Complexity
19	I think changing authentication mechanisms causes too much friction for users	Warranty — Complexity
20	I must use passwords at my workplace	Utility — Compatibility
21	I think there is an inconsistent experience across sites, browsers, and operating systems	Utility — Compatibility
22	I am concerned vendors strong-arm the use of their passkey management system	Utility — Compatibility
23	I worry that passkeys are not universally available	Utility — Compatibility
24	I am concerned about the reliance on a password manager for cross-device use	Utility — Compatibility

Study participants are collectively referred to as the P-set [4]. For this study, we recruited the P-set in April 2025 through the Prolific platform[5], applying filters to select individuals who use technology daily in their work and interact with a device every day. These criteria attempted to ensure that participants would be technically proficient and passkeys would be relevant and viable within their typical workflows. We deliberately adopted a broadly inclusive design for the P-set, as our primary objective was to examine passkey adoption across a general user population.

The Q-sort was administered online using the EQ Web Sort[6] tool and followed a multi-step procedure, as illustrated in Fig. 1. After providing informed consent,

[5] https://www.prolific.com/
[6] https://github.com/shawnbanasick/eq-web-sort

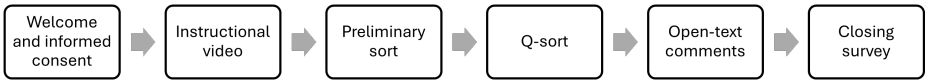


Fig. 1. Q-sort process

participants viewed a brief introductory video designed to (re)familiarise them with passkeys. Participants then completed a preliminary review of the statements by sorting them into three categories — disagree, no strong feelings, and agree. This step enables participants to become familiar with the Q-sample and makes the subsequent Q-sort more manageable. Participants next completed the main Q-sort activity, during which all statements were ranked within a provided grid according to their views. Following the Q-sort, participants provided open-text comments on the statements they most strongly agreed or disagreed with, offering additional insight into the reasoning behind their rankings. Finally, participants completed a short survey capturing their attitudes toward technology adoption, whether they held a technical role, and their current knowledge/use of passkeys. Completing the Q-sort required a median of 20 minutes, and participants were compensated at the equivalent of the UK National Living Wage.

Analysis of the Q-sorts involved correlation and factor analysis. In Q factor analysis group participants with similar viewpoints into factors based on their Q-sort configurations. Each factor therefore represents a distinct perspective on the topic. The open-text comments and survey responses provided supplementary context to support the interpretation of these factors [22,4,16]. All analyses were conducted using the specialised KADE[7] tool.

4 Analysis and Findings

We collected 40 responses from a demographically diverse sample. Participants ranged in age from 19 to 64 years (M = 37.83), with equal representation of male (n = 20) and female (n = 20) participants. Respondents resided in multiple countries, with the majority based in South Africa (n = 15), the United Kingdom (n = 14), and the United States (n = 8), and single participants from Chile, Italy, and Poland. Most participants were employed full time (n = 31).

The sample was predominantly self-described as technical (n = 32). In terms of technology adoption, participants most commonly identified as early adopters, either reporting that they prefer to be among the first to try new products, technologies, or ideas (n = 13), or that they adopt new products after observing initial evidence of success (n = 13). Passkey usage varied across the sample, with participants most frequently reporting infrequent use (n = 15) or use whenever possible (n = 14).

[7] https://github.com/shawnbanasick/kade

Diversity within the P-set is desirable to ensure representativeness and to maximise the likelihood that the important factors emerge [4]. Our study participants exhibited diversity in age, gender, and passkey usage. We also consider it essential to include participants who have adopted passkeys, as their perspectives on broader adoption reflect lived experience.

Factors were extracted using the centroid method. Three factors were retained for further analysis based on inspection of the scree plot and the criterion of eigenvalues exceeding 1.00. The retained factors were rotated using a varimax procedure, followed by a manual rotation of Factors 1 and 2 by -25 degrees to enhance interpretability. Factor 3 was split due to its bipolar structure. All factors selected for interpretation included at least two participants with significant loadings, consistent with recommended practice [25]. In this section, Pi refers to a comment by Participant i. Where statements and their rankings are referenced, the format (*Id*: *ranking*) is used to indicate how a statement (as listed in Table 1) was ranked on a 7-point scale ranging from strong disagreement (-3) to strong agreement (+3).

4.1 Factor 1: Satisfaction with the status quo

Factor 1 comprised 11 participants with significant loadings (4M/7F) and a mean age of 35.73 years. This factor accounted for 19% of the total variance, with an eigenvalue of 7.78 and a composite reliability of 0.98.

Most participants associated with this factor reported prior experience using passkeys. Specifically, four participants indicated that they used passkeys whenever possible, while an additional four reported infrequent use. As P11 explained, *"I believe that they are easy and convenient to use and save time trying to remember difficult passwords"*.

Despite passkeys experience, the highest-ranked statements for this factor reflected a preference for password managers (3: +3) and the use of MFA (12: +3). These rankings indicate a continued reliance on passwords, coupled with strategies to strengthen password-based authentication through secure management and additional protective layers. As P12 noted, *"combining methods feels more secure to me"*. Such views align with prevailing authentication guidance and best practices.

In contrast, the lowest-ranked statements indicated limited concern about the use of biometrics (10: -3) or the privacy of biometric data (9: -3). Similarly, trust in service providers was not perceived as problematic (11: -2). Although such issues are commonly cited as barriers to passkey adoption, they were not salient concerns for participants associated with this factor. Compared to other factors, these participants more strongly endorsed the view that passwords remain sufficient (2: +2) and expressed relative satisfaction with the current authentication landscape (1: +1).

Participants expressed ambivalence regarding the friction involved in transitioning to new authentication mechanisms (19: 0) and reported ongoing reliance on passwords in workplace settings (20: +2). As P3 observed, *"changing authentication mechanisms can cause too much friction for users because [of] the*

disruption of established habits and familiarity with existing systems". For participants aligned with this factor, password managers, particularly when combined with MFA, were viewed as an effective and satisfactory solution. As P17 explained, they were preferred *"because they are easy to use and safe"*, while P16 remarked, *"I am still happy not to use passkeys as for me MFA is pretty good"*. Overall, this factor reflects a pragmatic preference for maintaining the existing authentication status quo.

4.2 Factor 2: Concern about risk and loss of control

Factor 2 comprised eight participants with significant loadings (4M/4F) and a mean age of 44.63 years. This factor accounted for 11% of the total variance, with an eigenvalue of 4.59 and a composite reliability of 0.97.

Most participants associated with this factor again reported prior experience using passkeys. Three participants indicated that they used passkeys whenever possible, while a further three reported infrequent use. As P33 remarked, *"I think they are very convenient, and I hope more people start using them. I would also like to see my workplace migrate to them"*.

The highest-ranked statements for this factor reflected concerns about authentication fallback mechanisms, both when passkeys fail (6: +3) and when cross-device access depends on a password manager (24: +3). Participants expressed anxiety that such dependencies could introduce a single point of failure. As P2 explained, *"I am concerned that this can create a single point of failure"*, while P33 noted, *"I worry that I could be completely locked out of my accounts"*. The risk of losing access to services or accounts, potentially through no fault of the user, was therefore a salient concern for participants aligned with this factor.

The lowest-ranked statements indicated strong disagreement that passwords (2: -3) or single sign-on (4: -3) are sufficient authentication mechanisms. Accordingly, participants expressed dissatisfaction with the current authentication status quo (1: -1). Passkeys were not viewed as difficult to understand (17: -1) or to use (18: -2). As P13 observed, *"Although I have reservations in using a passkey, I do not like the default system of passwords. It is way too cumbersome"*.

In contrast to Factor 1, participants associated with this factor believed that changing authentication mechanisms introduces excessive user friction (19: +2) and reported lower levels of trust in service providers (11: +2). They also expressed concern about the privacy of biometric data used with passkeys (9: +1). As P22 cautioned, *"There are no guarantees on what my data will be used for and how secure it will be — hackers always seem to find a way"*. Overall, this viewpoint reflects a cautiously optimistic stance toward passkeys. As P2 summarised, *"Passkeys have the potential to revolutionize authentication, but it is important for us to be aware of the challenges and ongoing development"*. For these participants, the primary barrier to adoption was concern about single points of failure and the potential loss of access to accounts and personal data.

4.3 Factor 3

Factor 3 exhibited both positive and negative significant loadings and was therefore identified as a bipolar factor. To support interpretation, it was divided into two distinct factors [25], hereafter referred to as Factors 3a and 3b. Collectively, Factor 3 accounted for 7% of the total variance, with an eigenvalue of 2.77.

Factor 3a: Fragmented user experience (UX). Factor 3a comprised five participants with significant loadings (3M/2F) and a mean age of 41.4 years, and demonstrated a composite reliability of 0.95. This factor is characterised by participants who reported infrequent use of passkeys.

For this factor, the highest-ranked statements reflected frustration with the continued requirement to use passwords in workplace contexts (20: +3) and the necessity of relying on passwords even on services that support passkeys. These rankings suggest dissatisfaction with the persistent entrenchment of passwords in everyday authentication workflows. Participants also expressed concern about the limited availability of passkeys across services (23: +1) and viewed inconsistent passkeys experiences across websites, browsers, and operating systems as problematic (21: +2).

The lowest-ranked statements indicated relative comfort with how passkeys would handle multiple devices (13: -3), newly added devices (8: -2), and lost or stolen devices (7: -2). Participants also reported minimal concern regarding the use of biometrics (10: -3). Passkey terminology was not perceived as difficult to understand (17: -1), nor were passkeys considered difficult to use (18: -2). As P40 noted, *"I don't have problems with user interfaces – I enjoy learning them"*. Overall, participants associated with this factor appeared conceptually comfortable with passkeys and their underlying operation.

Participants expressed neutrality toward the use of MFA (12: 0) and toward concerns about fallback mechanisms when passkeys fail (6: 0). Instead, they expressed greater concern about being compelled by vendors to adopt proprietary passkey management systems (22: +1). Taken together, these responses suggest that for this group the user experience and extent of vendor control were more salient than security risks per se.

Factor 3b: Limited technical understanding. Factor 3b comprised three participants with significant loadings (3F) and a mean age of 32.33 years, with a composite reliability of 0.92. Participants associated with this factor were evenly divided between infrequent use, having experimented with passkeys but subsequently discontinuing their use, and having no prior experience with passkeys.

For this factor, the highest-ranked statements reflected uncertainty about how passkeys would handle lost or stolen devices (7: +3), newly added devices (8: +2), and, to a lesser extent, multiple devices (13: 0). Participants expressed concern about potential unauthorised access and recovery processes. As P31 noted, *"I feel like someone could just grab my phone and access my computer"*, while P25 stated, *"I'm unsure how would I be able to de-attach a former device*

and authenticate my access using a completely new device". These responses suggest limited familiarity or comfort with the technical details underlying passkey systems. Consistent with this interpretation, participants indicated difficulty understanding passkey-related terminology (17: +1) and discomfort with the use of biometrics (10: +1).

In contrast, the lowest-ranked statements indicated that participants did not perceive passkey setup as difficult (15: -3), nor were they concerned about the continued requirement to use passwords in workplace contexts (20: -3) or on services that support passkeys (5: -2).

Notably, participants associated with this factor did not view password managers as meaningfully reducing the burden of password management (3: -2). Instead, they regarded single sign-on as a preferable alternative (4: +3) and expressed a modest degree of satisfaction with the current authentication status quo (1: +1). Overall, the primary barrier for these participants appears to be uncertainty about device handling and difficulty understanding passkeys terminology, rather than concerns about the user experience or password persistence.

5 Discussion

Our findings identify four shared perspectives on passkeys. A primary factor limiting the adoption of passkeys appears to be users' satisfaction with existing password-based practices. This position does not necessarily imply poor security hygiene. Rather, many participants reported maintaining strong security through the use of password managers and MFA. Although passkeys were generally viewed as a viable alternative, participants did not perceive a sufficiently compelling incentive to transition from established methods. This reluctance may, in part, stem from concerns about learnability. Prior work suggests that authentication schemes must be easily learnable and memorable for users unfamiliar with them [3]. While most participants described themselves as technically competent, they expressed reservations about the ability of less technically experienced users to adopt passkeys successfully. As P16 noted, *"I personally think passkeys are fine . . . because I am knowledgeable in this area. I think an average user would struggle to understand the concept"*. Similarly, P3 raised concerns about accessibility and inclusivity, pointing to potential barriers for users without newer devices or sufficient technical expertise.

A second important factor concerns apprehensions about potential failures within the passkey authentication workflow. When coupled with lower levels of trust in service providers, these technical concerns act as a barrier to adoption, even among users who report dissatisfaction with current password-based approaches. As P2 noted, *"I worry that it could create a vendor lock-in, making it difficult to switch users in the future. I am also concerned about the risk of security breaches or data leaks if the vendor is compromised"*. These concerns reflect less scepticism toward the concept of passkeys itself and more uncertainty about implementation practices and the assurances that vendors can provide regarding correct and secure operation. Echoing this sentiment, P38 stated, *"I*

think things can often go wrong, and I do worry about how my data is being handled". Overall, this viewpoint remains generally positive toward passkeys but highlights perceived challenges and the need for further technical maturation before widespread adoption is considered viable.

The final viewpoint we identified reflects a dual emphasis on user knowledge and an acceptable user experience. Without a sufficient understanding of how passkeys function, including the associated terminology, users are likely to be hesitant to adopt them. Addressing this barrier will require user education tailored to varying levels of technical competence. As P24 observed, *"Everyone's level of knowledge and knowhow when it comes to device security is different, so it's not a one size fits all solution"*. However, even when users possess adequate conceptual understanding, a positive and seamless user experience remains essential for adoption. Prior research has suggested that digital nudges introduced at key interaction points may support passkey uptake [20]. Our findings extend this work by offering empirically grounded guidance on how such nudges might be framed to address user concerns and expectations.

Notably, two statements elicited relative consensus across all factors, indicating viewpoints shared by all participants. The first reflected disagreement with the notion that passkey user interfaces are difficult to use (18: -2/-1). This suggests that the overall user experience across the authentication workflow may be more influential than the design of any specific user interface element. The second indicated relative neutrality and slight disagreement with the claim that roaming authenticators are difficult to use (14: 0/-1). This may highlight a specific area in which targeted user education could support improved understanding and adoption.

6 Conclusion

This study used Q-methodology to investigate why users are not adopting passkeys. It surfaced four distinct viewpoints from 40 technically proficient daily technology users.

The findings reveal that passkey non-adoption is not monolithic but spans four distinct stances. Factor 1 participants view their current password-based practices (augmented by password managers and MFA) as sufficiently secure and convenient. Factor 2 participants are broadly supportive of passkeys in principle but are held back by concerns about single points of failure, loss of access, and insufficient trust in vendors to guarantee correct and secure operation. Factor 3a participants are conceptually comfortable with passkeys but frustrated by fragmented user experience across platforms, browsers, and operating systems, and by vendor control over the passkey ecosystem. Factor 3b participants lack sufficient understanding of how passkeys handle core scenarios such as device loss and replacement, and find the associated terminology unclear.

Framing authentication as a service and applying the ITIL utility/warranty lens yield additional interpretive clarity: For Factor 1, passwords' warranty failures have been sufficiently patched through coping mechanisms such that passkeys

may offer no compelling incremental value. Factor 2's concerns map primarily onto warranty — these users question fitness for use, not fitness for purpose. Factors 3a and 3b raise distinct utility concerns: whether the service works consistently across contexts, and whether users can understand what it does.

Because these viewpoints are heterogeneous, no single intervention will suffice. Passkey advocates must strengthen the value proposition relative to password managers and MFA, improve the transparency and robustness of account recovery and fallback mechanisms, achieve greater UX consistency across platforms, and provide clearer terminology and educational resources for less technically confident users.

It is important to emphasise that this analysis identifies viewpoints i.e., patterns in how participants perceive and interpret passkeys. Although the relatively small number of participants precludes claims about capturing all possible viewpoints, this sample size is methodologically appropriate for the identification of shared perspectives [25]. While the viewpoints presented here are valid and empirically grounded, it should be acknowledged that additional perspectives on passkey adoption may exist beyond those identified. Our aim, consistent with the philosophical foundation of Q, was to identify the presence of distinct viewpoints rather than to generalise [4]. Accordingly, we make no claims regarding the proportion of the population that aligns with any given viewpoint.

Future work should extend the investigation to less technical user populations, track viewpoint shifts longitudinally as the ecosystem develops, and use the identified typologies to design and evaluate targeted adoption interventions.

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